

RESUME

RESUME — HelixLLM Full-Extension Programme (session-resumption file, §11.4.131)

Fresh session? Paste the SHORT prompt below. This file is the single out-of-the-box entry point. Always current; re-read on every new session. Track (T1/main).

SHORT resumption prompt (paste into a fresh session)

Read docs/research/07.2026/00_master/RESUME.md + .superpowers/sdd/progress.md (the live SDD ledger), run `git fetch --all`, and continue the HelixLLM full-extension programme. Phase 0 (GPU) + Phase 1 (fleet + fixes) + Phase 2 (HelixAgent → HelixLLM e2e) are PROVEN; the LLMsVerifier capability-verification chain (C1 → C2 → C4 → C5 → C3) is fully landed and under a final combined independent review. Next: land the review to GO, then the release-prep pass (pointer bumps + prefixed release tag §11.4.151 via merge-onto-latest §11.4.113, NO force-push). Honor anti-bluff §11.4, subagent-driven §11.4.70, (T1/<branch>) labels §11.4.182, one canonical branch feature/helixllm-full-extension §11.4.181, §11.4.174 shared-host process/tree ownership.

Current phase + immediate next action

- **Phase 0 (GPU foundation):** COMPLETE — rootless CDI passthrough + sm_120 build + real 30B inference PROVEN.
- **Phase 1 (fleet + fixes):** 30B coder live; Containerfile + claude_toolkit fixes landed, all re-reviewed GO.
- **Phase 2 (HelixAgent → HelixLLM e2e):** PROVEN + review GO — real generate + Postgres/Redis persistence (cognee/vector honest SKIP, OQ2).
- **LLMsVerifier chain:** C1 C2 C4 C5 C3 + advisories all landed, combined review GO — release-ready.
- **Phase-3:** embeddings IMPL PROVEN (55bdf9b6, real bge-small TEI, cos margin 0.3578) · VRAM broker CORE GO (a12df57c — unblocks GPU tiers)

- designs done: embeddings (cf26b813), translation (c9ac8683), provider-coverage (1e6f3347, 0 new wire adapters).
- **Immediate next (fresh session):** (a) the pending §11.4.142 reviews are DONE for broker; run reviews of the 3 design docs + embeddings if desired; (b) GPU tiers now unblocked — vision (VLM)/image/video gen via the broker; (c) provider config-rollout (13 providers, config-only); (d) translation IMPL (design c9ac8683); (e) cognee P-OQ2-A wire; (f) small follow-ups: OQ1 doc correction in PHASE2_BLOCKERS_INVESTIGATION.md, P3-EMB-1 golden-good dim-aware fixture, detector.go:61; (g) release-prep + prefixed tag when scope-complete.
- **Terminal goal (this scope):** a fully-validated, prefixed release tag (§11.4.151) published across main + all owned submodules; local HelixLLM on the RTX 5090 exposed via HelixAgent to HelixCode/CLI agents.

LIVE SERVER (operator-testable, running now)

Qwen3-Coder-30B-A3B serving on `http://localhost:18434/v1` (OpenAI-compatible) — container `helixllm-coder` (`podman ps`), `--network=host`, image `localhost/helixllm/llamacpp-router:cuda12.8-sm120`, 8 parallel slots, 24k ctx, q8_0 KV, ~19.4 GB VRAM. PROVEN: single-stream ~220 tok/s (`coder_live_e2e` log; RESUME's older ~322 tok/s was a different `-jinja/q8_0-KV` run) + 8 concurrent agents @85–96 tok/s; real coding output (`is_palindrome`, `func Add`). Restart: `podman start helixllm-coder`.

ENDPOINT GOTCHA (load-bearing, proven in Phase-2 docs/qa/phase2_e2e_20260706/12_endpoint_finding.txt): raw `curl` uses `http://localhost:18434/v1/chat/completions`. A client/SDK that APPENDS `/v1/chat/completions` (incl. HelixAgent via `HELIX_LLM_LOCAL_OPENAI_ENDPOINT`) MUST use the BASE `http://localhost:18434` (NO `/v1`) — else `.../v1` → double `/v1/v1` → **HTTP 404**. (This SUPERSEDES the `:18434/v1` value in `10_llmsverifier_helixagent/PHASE2_BLOCKERS_INVESTIGATION.md` OQ1 — that doc line is pending a correction commit in the release-gate sweep.) NOTE: latest `llama.cpp` `-fa` takes a value (`-fa on`).

Live-state anchors (facts, §11.4.6)

Anchor	Value
Constitution HEAD followed	0882b9e (through §11.4.182)
Host	ALT Workstation 11.1; RTX 5090 32 GB; driver 570.169; CUDA 12.8;

Anchor	Value
	podman 5.7.1 rootless; 64 cores / 251 GiB
Canonical branch (§11.4.181)	feature/helixllm-full-extension — ACTIVE (no upstream tracking configured yet)
Router image (built+proven)	localhost/helixllm/llamacpp-router:cuda12.8-sm120 — latest llama.cpp, sm_120, OpenSSL/curl (-hf HTTPS proven), ships rpc-server
Release prefix (§11.4.151)	HELIX_RELEASE_PREFIX else helix_code
Shared-host caution (§11.4.174)	helix_agent checkout carries a CONCURRENT QA/dep track's uncommitted go.mod/go.sum + .qa_bak — NOT ours; do not sweep on pointer-bump

Done so far this session (real, evidence-backed commits)

- helix_llm 13d2d27 Containerfile: hard-fail ggml-rpc-server copy (§11.4.122) + OpenSSL/curl — re-review **GO**; d8b3fa2 phase-1 QA evidence (-hf 469MB HTTPS download + live coder e2e); 3f85e3d5 OPERATOR_GUIDE.
- claude_toolkit ef77b19 loud-fail resolution + discriminating test; 9d12347 C1 regression guard (37/0, §1.1 mutation-proven) — **REVIEW-3 CLEAN**.
- llms_verifier 09f9533c C4 · 28e6625a C5 · ad18e91f C3 · d1f04e5c advisories (detector gate + sentinel oracle) — **chain + advisories complete, combined review GO**.
- helix_llm a12df57c **VRAM broker CORE** (Budget/admission/single-owner, review GO — unblocks GPU tiers).
- helix_agent 17f08ba9 Phase-2 live tests (provider e2e + redis) — pointer NOT yet bumped (§11.4.174).
- helix_code cf26b813 embeddings design · 9bf4c3da Phase-2 stack design · da0fabae blockers investigation · 5223d10d C3 handoff · 278df582 Phase-2 e2e evidence · c9ac8683 translation design · 1e6f3347 provider-coverage design · 55bdf9b6 **embeddings IMPL proven** (bge-small TEI, cos margin 0.3578).

Next actions (in order)

1. Land the combined **C4+C5+C3 review** to GO (adjudicate detector.go:61 no-probe self-cert).
2. Correct PHASE2_BLOCKERS_INVESTIGATION.md OQ1 verdict → base <http://localhost:18434>.
3. Release-prep: bump main-repo submodule pointers (helix_llm, claude_toolkit, llms_verifier; helix_agent carefully per §11.4.174) → §11.4.40 full-suite pre-tag sweep → prefixed tag (§11.4.151) → publish via merge-onto-latest (§11.4.113, NO force-push §2.1 multi-upstream).
4. Post-release follow-ups (tracked): P-OQ2-A wire cognee_memory_repository + P-OQ2-B re-verify cognee 1.2.2 bug; detector.go:61 no-probe self-cert; VRAM broker IMPL (design 5102607 done) to unlock GPU vision/image/video; Phase-3 CPU embeddings + translation NMT impl (designs done).

Binding constraints (do not violate)

Anti-bluff §11.4 (real captured proof, no metadata PASS) · runtime-signature = done §11.4.108 · no-force-push merge-onto-latest §11.4.113 · rootless podman via containers submodule §11.4.76/§11.4.161 · no CI §11.4.156 · no silent removal §11.4.122 / investigate-before-remove §11.4.124 · deep multi-angle research per change §11.4.150 · independent review to GO §11.4.125/§11.4.142/§11.4.134 · one canonical branch §11.4.181 · (T<N>/<branch>) labels §11.4.182 · shared-host ownership §11.4.174.